

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Mock Interview

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.*
(BL5)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 25

Code of Conduct

I G. Venkata Niranjan Reddy (192525199) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Mock Interview

Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Mock Interview Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Knowledge	25%	Shows deep and accurate technical understanding	Good technical understanding	Basic understanding	Weak understanding
Problem Solving	20%	Responds with strong architecture-level reasoning	Reasonable reasoning	Basic reasoning	Weak reasoning
Communication	20%	Answers are confident, precise, and professional	Mostly clear communication	Moderate clarity	Weak communication
Practical Awareness	20%	Connects concepts to real-world implementation well	Some practical relevance	Limited practical relevance	No practical linkage
Confidence and Professionalism	15%	Highly confident and professional	Generally professional	Moderately professional	Low confidence and professionalism

1. Explain the difference between HDFS and traditional file systems.

Answer:

HDFS, or Hadoop Distributed File System, is designed to store very large volumes of data across multiple machines in a cluster. A traditional file system usually stores data on a single server or a limited storage device.

The main difference is that HDFS divides large files into blocks and distributes those blocks across different Data Nodes. It also maintains multiple replicas of each block to provide fault tolerance. Traditional file systems generally depend on a single storage system and may require separate backup mechanisms.

From an architecture perspective, HDFS is optimized for high-throughput batch processing and scalability, whereas traditional file systems are typically optimized for low-latency access and individual file operations.

2. Why is replication important in distributed storage systems?

Answer:

Replication is important because hardware failures are common in distributed environments. If a server or disk fails, copies of the data stored on other nodes can still be accessed.

For example, in HDFS, a replication factor of 3 means that each data block is stored on three different Data Nodes. If one Data Node fails, the system can retrieve the block from another replica.

Replication therefore improves:

- Data availability
- Fault tolerance
- Reliability
- Recovery from hardware failures

Architecturally, replication prevents a single machine failure from causing permanent data loss or interrupting processing.

3. How does MapReduce divide work between map and reduce phases?

Answer:

MapReduce divides a large processing task into two main phases: Map and Reduce.

In the Map phase, the input data is divided into smaller splits, and multiple mapper tasks process these splits in parallel. Each mapper converts the input into intermediate key-value pairs.

During the Shuffle and Sort phase, the framework groups all values associated with the same key.

In the Reduce phase, reducer tasks process the grouped data and generate the final output.

For example, in a word-count application:

- Mapper: (document) → (word, 1)
- Shuffle: Groups all identical words
- Reducer: (word, [1,1,1...]) → (word, total count)

This architecture allows large datasets to be processed in parallel across many cluster nodes.

4. What role does YARN play in large-scale Hadoop processing?

Answer:

YARN, or Yet Another Resource Negotiator, is responsible for managing cluster resources and scheduling applications.

Its main components include:

- **Resource Manager** – Manages overall cluster resources.
- **Node Manager** – Manages resources on individual worker nodes.
- **Application Master** – Coordinates resources and tasks for a specific application.
- **Containers** – Provide allocated CPU and memory resources for tasks.

From an architecture perspective, YARN separates resource management from data processing. This allows different processing frameworks, such as MapReduce and Spark, to share the same Hadoop cluster efficiently.

5. Differentiate NAS and SAN in an enterprise data environment.

Answer:

NAS, or **Network Attached Storage**, provides file-level storage over a network. Users and applications access files using protocols such as NFS or SMB.

SAN, or **Storage Area Network**, provides block-level storage. Servers see SAN storage as if it were directly attached disks.

Feature	NAS	SAN
Storage Type	File-level	Block-level
Network	Standard IP network	Dedicated storage network
Performance	Moderate to high	Very high
Complexity	Easier to manage	More complex
Typical Use	File sharing and backups	Databases and enterprise applications

In enterprise architecture, NAS is generally suitable for shared file storage, while SAN is preferred for high-performance applications requiring low-latency block access.

6. What is the purpose of ETL in a big data pipeline?

Answer:

ETL stands for **Extract, Transform, and Load**.

- **Extract:** Collect data from sources such as databases, applications, sensors, and logs.
- **Transform:** Clean, validate, standardize, and convert the data into a suitable format.
- **Load:** Store the processed data in systems such as HDFS, data warehouses, or data lakes.

The main purpose of ETL is to convert raw and inconsistent data into reliable, usable data for analytics and decision-making.

From an architecture perspective, ETL acts as the bridge between distributed data sources and analytical platforms.

7. How does Apache NiFi help in data integration workflows?

Answer:

Apache NiFi is a data-flow automation platform that helps organizations collect, route, transform, and manage data between different systems.

It supports:

- Data ingestion from multiple sources
- Real-time data routing
- Data transformation
- Flow prioritization

For example, a large organization can use NiFi to collect application logs, database records, and IoT sensor data, transform them into a common format, and send them to HDFS or Kafka.

Architecturally, NiFi acts as a central data-flow layer that simplifies integration between distributed systems.

8. What are the key failure points in a Hadoop cluster and how would you handle them?

Answer:

The main failure points include:

1. **Data Node failure** – A worker node storing HDFS blocks becomes unavailable.
Solution: Use HDFS replication and automatically recreate missing replicas.
2. **Name Node failure** – The metadata service becomes unavailable.
Solution: Configure Name Node High Availability with an active and standby Name Node.
3. **Network failure** – Nodes cannot communicate properly.
Solution: Use network redundancy, monitoring, and automatic task rescheduling.
4. **Disk failure** – Storage devices become corrupted or unavailable.
Solution: Use replication, disk monitoring, and replacement of failed hardware.
5. **Resource Manager failure** – Cluster resource scheduling is interrupted.
Solution: Configure Resource Manager High Availability.

A reliable Hadoop architecture should assume that failures will occur and should use redundancy, replication, monitoring, and automatic recovery.

9. How would you design a secure data ingestion pipeline for a large organization?

Answer:

I would design the pipeline with security at multiple layers:

Data Sources → Secure Ingestion → Validation → Processing → Secure Storage → Analytics

First, data would be collected through authenticated and encrypted connections using TLS. Tools such as Apache NiFi or Kafka could be used for controlled ingestion.

Before storing the data, the pipeline should perform:

- Authentication and authorization
- Input validation
- Data quality checks
- Malware or suspicious data detection where applicable
- Encryption of sensitive information

Access to Hadoop resources should be controlled using role-based access control. Sensitive data should be encrypted both in transit and at rest.

The architecture should also include:

- Audit logging
- Monitoring and alerting
- Key management
- Data lineage
- Backup and disaster recovery

This creates a defense-in-depth architecture, where security is applied throughout the entire data lifecycle rather than only at the network boundary.

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Answer:

I would explain Hadoop as a system that allows an organization to process very large amounts of data by dividing the work across many computers.

Instead of purchasing one extremely powerful and expensive server, the organization can use multiple machines working together as a cluster. When the amount of data increases, additional machines can be added to increase storage and processing capacity.

For example, if the company processes customer transactions, application logs, and sensor data, Hadoop can distribute this workload across multiple servers and process it simultaneously.

The main business benefits are:

- Handles increasing data volumes
- Reduces dependence on a single large server
- Provides fault tolerance through replication
- Supports large-scale analytics
- Can scale by adding more nodes

In simple terms, Hadoop is like dividing a large project among many employees instead of asking one employee to complete everything alone. This improves scalability, reliability, and processing capacity.